Striking While the Iron Is Hot vs Steady Does It: Hours Worked over the Business Cycle and Welfare

Christian Jensen*
University of South Carolina

May, 2025

Abstract

To capitalize on exogenous changes in productivity, hours worked fluctuate over the business cycle. However, we find that the loss in welfare from holding aggregate hours constant over the cycle would at most be the equivalent of .12% of consumption, or $66 per person per year in the U.S. in 2023, for standard parameter values. This is remarkably low considering that our computations assume innocuous intensive-margin adjustments in hours, while in reality most of the fluctuations are in terms of costly cycles of layoffs and hiring involving unemployment, the loss of firm-specific human capital and time-consuming job market search.

JEL Codes: E32, E24
Keywords: Aggregate hours worked; Cyclical fluctuations

*Department of Economics; Darla Moore School of Business; University of South Carolina; 1014 Greene Street; Columbia, SC 29208; 803-777-2786; cjensen@alumni.cmu.edu.

1 Introduction

Hours worked fluctuate over the business cycle, presumably to strike while the iron is hot by working more when productivity is high and less when it is low, thus raising the number of goods obtained per hour of labor. At the same time, common utility functions imply a preference for smoothness that penalizes fluctuations in leisure. More importantly, most of the cyclical fluctuations in aggregate hours worked are along the extensive margin (Cho and Cooley (1994)). Hence, they involve layoffs that lead to spells of unemployment and the loss of firm-specific human capital (Becker (1964), Topel (1990) and Lazear (2009)), as well as hiring that requires both employers and workers to undertake time-consuming search, screening and training (Okun (1963), Krueger and Mueller (2010) and Blatter et al (2012)). One might expect the rewards for enduring such costly adjustments to be correspondingly large. However, even when substituted with innocuous intensive-margin adjustments, we find the impact on welfare from shutting down cyclical fluctuations in aggregate hours to be trivially small. At most the equivalent of .12% of steady-state consumption, or \$66 per capita per year in the U.S. in 2023, according to a standard real business cycle model in the tradition of Kydland and Prescott (1982). Even when pushing the calibration to the extreme, the cost remains below .5%, or \$264.

That firing and hiring can be costly is evidenced by firms hoarding labor over the business cycle to mitigate such expenses (Fair (1985), Fay and Medoff (1985)). But why then are not all the cyclical movements in hours along the intensive margin? Individual workers have incentives to defend their own hours and paychecks. Employers may be better off retaining only the most productive, enticing them with stable hours and pay. The prevalence of programs that compensate for job loss, rather than reductions in hours, may also play a role. Moreover, social norms dictate full-time employment. Hence, in practice, most of the variation in aggregate hours are extensive margin (Cho and Cooley (1994)). However, while employers may have no choice but to resort to layoffs to survive downturns, our computations show that the cost to the economy from eliminating such cyclical variations, for example through temporary wage subsidies as those introduced during the COVID-19 pandemic to limit furloughs and layoffs (Eichhorst et al. (2022)), would be very modest.

Eliminating the cyclical fluctuations in employment should be less costly than eliminating those in hours, since it should be feasible to vary hours with-

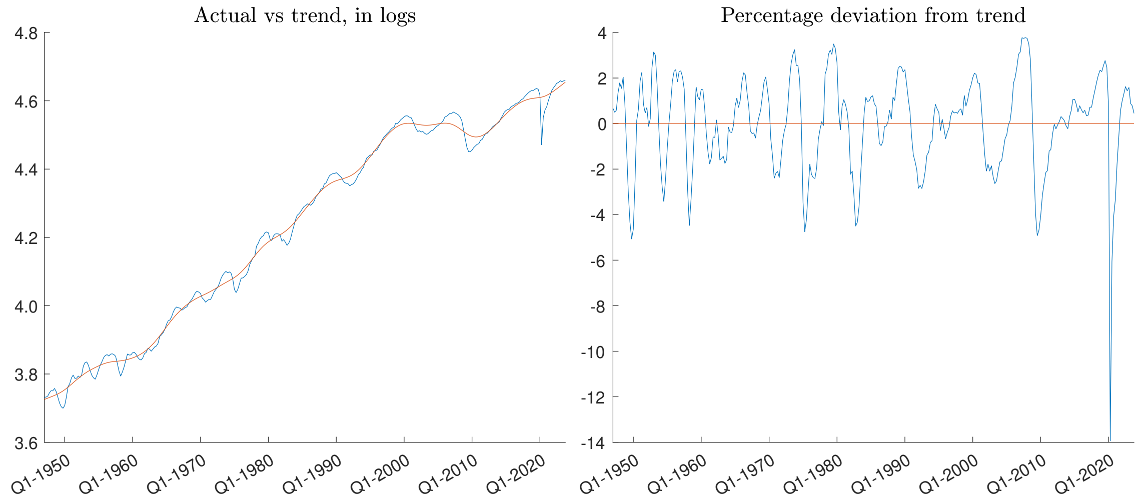

Figure 1: Aggregate hours worked in the nonfarm business sector in the U.S. 1948Q1-2023Q4. Trend computed with the HP-filter.

out changing the number of employees, for instance using overtime, to exploit changes in productivity. A precise estimate would require a model that distinguishes between variations along the intensive and extensive margins, and incorporates all the costs associated with hiring and firing. It should also allow employers the option to hoard labor to mitigate the impact on their bottom line. Rather than taking on such a monumental task, we show that the cost of eliminating all cyclical fluctuations in aggregate hours is trivially small, even assuming that these were exclusively innocuous intensive margin adjustments. Clearly, incorporating savings of costs associated with hiring and firing would lower the estimated net welfare cost. Hence, the only way it could change our conclusion is if it made the net welfare cost go from trivially small to negative. Due to the lack of precision, the guiding principle throughout this study is erring on the side of caution, overestimating rather than underestimating the cost of holding aggregate hours constant over the business cycle.

Figure 1 shows aggregate hours worked in the nonfarm business sector in the U.S. between 1948 and 2023. The left-hand-side panel plots the logarithm of the values in blue, together with the trend in red, obtained with the HP-filter (Hodrick and Prescott (1997)). The panel on the right shows the percentage deviations from trend, that is, the cyclical fluctuations. The present study computes the impact on welfare from adhering to the red lines instead of the blue, in an economy experiencing standard productivity shocks. However, this is not carried out for the single realization of the historical U.S. data plotted, but for a model calibrated to generate cycles with similar properties as those observed for the U.S. economy. By using the HP-filter to calibrate the cyclical fluctuations in hours that are shut down, only short-term variations are smoothed out, not the impact of structural change, growth of the labor force, and other long-term trends, which remain in the red line in the left-hand-side panel of figure 1. We focus on aggregate hours in the nonfarm business sector because it is the most volatile measure of hours at business-cycle frequencies, resulting in the highest calibrated variance for the underlying productivity shocks, and thus the largest possible impact from striking while the iron is hot. This translates into the highest possible cost of keeping hours constant despite productivity fluctuating.

The more elastic the labor supply, and the lower risk aversion (or higher intertemporal substitution), the more inclined the economy is to varying hours with productivity, raising the gains from doing so. It is therefore desirable to use a utility function that permits controlling the risk aversion and labor supply elasticity. Among the functions that allow for this, the one introduced by MaCurdy (1981) yields the highest cost of maintaining hours constant, so we focus on it. Results for the more popular King-Plosser-Rebelo (KPR henceforth, see King et al. (1988)) and Greenwood-Hercowitz-Huffman (GHH henceforth, see Greenwood et al. (1988)) preferences are also provided (with the appendix going through the details, see Boppart and Krusell (2020) for a discussion of these utility functions).

The present study builds upon the literature on the welfare cost of business cycles, initiated by Lucas (1987). Very briefly, it finds that, barring extreme risk (Barro (2009)) or risk aversion (Tallarini (2000), Otrok (2001)), the costs of fluctuations are miniscule, or even slightly negative (Krusell and Smith (1999), Gomes et al. (2001), Lucas (2003)), except when factoring in unemployment, especially with scarring (Storesletten et al. (2001), Krusell and Smith (1999), Krusell et al. (2009)). Those costs are computed by comparing expected lifetime utility with and without stochastic shocks, the latter yielding an economy where consumption and hours are both constant over time. In contrast, the present study computes the cost associated with maintaining just hours constant, in the presence of the aggregate productivity shocks believed to be the main driver of business cycles (Kydland and Prescott (1982) and Aiyagari (1994)).

The next section introduces a standard neoclassical dynamic stochastic general equilibrium real business cycle model. It also describes the use of large-scale grid-search methods to compute the impact on welfare from holding hours worked constant over the business cycle. The subsequent section presents the results, and shows that the computed cost of constant hours remains small for different utility functions and parameter values. Finally, we offer some concluding remarks. An appendix provides additional details and instructions for replicating the results.

2 Model

Imagine a continuum of measure one of identical households seeking to maximize expected discounted lifetime utility, solving

$$v(k_0, z_0) = \max E_0 \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \tag{1}$$

with preferences as proposed by MaCurdy (1981), so that

$$u(c_t, n_t) = \frac{c_t^{1-\varepsilon}}{1-\varepsilon} - \psi \frac{n_t^{1+1/\theta}}{1+1/\theta} \tag{2}$$

with respect to consumption $\{c_t\}_{t=0}^{\infty}$, hours of work $\{n_t\}_{t=0}^{\infty}$ and physical capital $\{k_{t+1}\}_{t=0}^{\infty}$, subject to the resource constraint

$$c_t = e^{z_t} A k_t^{\alpha} n_t^{1-\alpha} + (1-\delta)k_t - k_{t+1} \tag{3}$$

and the exogenous law of motion for productivity

$$z_t = \rho z_{t-1} + e_t \tag{4}$$

given initial capital $k_0 > 0$ and productivity z_0. The shock e_t is Normally distributed white noise with mean zero and standard deviation $\sigma > 0$, while the persistence parameter $\rho \in (0,1)$. The relative risk aversion coefficient (RRAC) for consumption $(-u''_{cc}c/u'_c)$ is $\varepsilon \geq 0$, the Frisch elasticity of labor supply is $\theta > 0$, the discount rate $\beta \in (0,1)$, the weights $\psi > 0, A > 0, \alpha \in (0,1)$, while the depreciation rate $\delta > 0$.

We use the first-order condition for hours of work

$$\psi n_t^{1/\theta} c_t^{\varepsilon} = (1-\alpha) e^{z_t} A k_t^{\alpha} n_t^{-\alpha} \tag{5}$$

	RRAC									
	0	.5	1	2	3	4	5	10	15	20
σ	.0154	.0486	.0610	.0853	.1013	.1178	.1310	.1375	.1400	.1400

Table 1: Volatility σ necessary for simulated standard deviation of HP-filtered consumption and hours to be at least as high as in U.S. data. MaCurdy preferences.

together with the resource constraint (3), to solve numerically for the optimal n_t for each possible combination of z_t, k_t and k_{t+1}. This permits substituting out for n_t, reducing the households' maximization to finding the optimal k_{t+1} for every possible combination of z_t and k_t. The model is solved numerically through value-function iterations using a grid search over a discretized version of the capital choice (see Ljungqvist and Sargent (2018)). This preserves the impact uncertainty has on risk-averse households' utility-maximizing behavior, including precautionary saving. We use 4 000 gridpoints for capital k and 51 for the exogenous productivity shock z, which is discretized using a Markov chain.[1] We prioritize k over z because the latter is exogenous. A fine grid for k is crucial to get as close as possible to the truly optimal solution.[2] As is shown in the appendix, our results are not very sensitive to the number of gridpoints used, so going above 4 000 or 51 is unlikely to have much impact, but if anything, would reduce the computed impact on welfare from keeping hours unchanged.

When holding hours constant, they are set equal to the steady-state level n_{ss}, with

$$n_t = n_{ss} \tag{6}$$

replacing the first-order condition (5). This may not be the optimal constant level in all situations, but computing that optimal level is a non-trivial task. While using the steady-state value can exaggerate the cost of maintaining hours unchanged, the computed cost is still trivially small.

[1]See the appendix and *markovappr.m* in Miao (2020) for details.

[2]The first-order condition for hours of work (5) presumes a continuous adjustment of both capital and hours. Therefore, using it to avoid expanding the dimensionality of the grid search requires having sufficient gridpoints for capital so that being constrained to a grid does not affect the optimal choice of hours. When the grid is too coarse, it can be optimal to deviate from the first-order condition (5) to compensate for not being able to adjust capital continuously. The appendix shows that the results do not vary significantly with the number of gridpoints when it is 1 000 or larger.

For given steady-state capital k_{ss} and hours n_{ss}, we derive the corresponding

$$A = \frac{1/\beta - 1 + \delta}{\alpha k_{ss}^{\alpha-1} n_{ss}^{1-\alpha}} \tag{7}$$

from the steady-state version of the first-order condition with respect to capital,

$$c_{ss} = A k_{ss}^{\alpha} n_{ss}^{1-\alpha} - \delta k_{ss} = (1/\beta - 1 + (1-\alpha)\delta)\alpha^{-1} k_{ss} \tag{8}$$

from the steady-state version of the budget constraint (3), and ψ from the steady-state version of the first-order condition with respect to hours (5). In our baseline, we follow the real business cycle literature and set the Frisch labor supply elasticity $\theta = 2$. In addition, $k_{ss} = 100, n_{ss} = 1/3, \alpha = 1/3, \beta = .99, \delta = .025$ and $\rho = .9$, which yields $A = 4.72$ and $c_{ss} = 8.03$. We vary the RRAC $\varepsilon \in \{0, .5, 1, 2, 3, 4, 5, 10, 15, 20\}$, and adjust the standard deviation of productivity shocks σ for each case so that HP-filtered consumption and hours are both at least as volatile as in the U.S. data over the period 1947Q1-2023Q4 (see the appendix for details). The resulting values for σ from the model with variable hours, provided in table 1, are used also in the counterfactual with constant hours. These σ-values are large compared to measurements from Solow residuals, and what is common in modeling, more so the higher the RRAC. However, because our focus is on utility, it is more important to match the volatility in consumption and hours than that of output and Solow residuals. Using lower values for σ would reduce the computed cost of holding hours constant (see appendix).

While arbitrary, the normalization $n_{ss} = 1/3$ and $k_{ss} = 100$, has no impact on the results. It does affect the scaling of the variables, and therefore lifetime utility $v(k_0, z_0)$. To neutralize for this, and the curvature of the utility function, we follow Lucas (1987) and compute the permanent percentage change in steady-state consumption required to generate an equivalent change in expected lifetime utility in an identical economy without uncertainty ($\sigma = 0$). Hence, if keeping hours constant makes expected lifetime utility $v(k_0, z_0)$ change by Δv, we set

$$\sum_{t=0}^{\infty} \beta^t u\left(c_{ss}\left(1+x\right), n_{ss}\right) - \sum_{t=0}^{\infty} \beta^t u\left(c_{ss}, n_{ss}\right) = \Delta v \tag{9}$$

to obtain the equivalent change

$$x = \left(1 + (1-\beta)(1-\varepsilon) c_{ss}^{\varepsilon-1} \Delta v\right)^{\frac{1}{1-\varepsilon}} - 1 \tag{10}$$

RRAC									
0	.5	1	2	3	4	5	10	15	20
.0383	.0860	.0735	.1169	.2542	.5242	.9048	2.605	4.501	6.346

Table 2: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of steady-state consumption. MaCurdy preferences, $gp = 4\,000, gpz = 51$.

in non-stochastic steady-state consumption c_{ss}. Since $\Delta v < 0$, $x < 0$, so we report the cost of keeping hours constant as $-x \times 100\%$, that is, the equivalent percentage drop in consumption relative to if hours had been allowed to vary freely. The computed value function $v(k_0, z_0)$ is evaluated at steady-state capital and productivity, setting $k_0 = k_{ss}$ and $z_0 = 0$. Results are similar if evaluating instead at the average capital ($k_0 = \bar{k}$), and also as a fraction of the average consumption pertaining over the business cycle with stochastic shocks (see appendix).

3 Results

Table 2 shows the computed cost of keeping aggregate hours constant over the business cycle as a percentage of steady-state consumption, for different RRACs, when all other parameters are at their baseline. As a guide, a meta-analysis by Elminejad et al. (2022) based on 1 021 estimates of the RRAC from 92 studies, yields a mean estimate of 1. Moreover, Chetty (2006) argues that a coefficient above 2 is inconsistent with observed labor supply behavior. For a RRAC of 2 or below, as is typical in macroeconomic models, the cost is at most .12% of steady-state consumption, or \$66 per capita in the U.S. for 2023 (given per capita consumption expenditures of \$56 202). The computed cost is higher for RRACs above 2, going beyond 1% of consumption for RRACs above 5. For RRACs ≥ 2, the cost is increasing in risk aversion when holding the underlying volatility σ constant. But with MaCurdy preferences, it is necessary to raise σ with the RRAC in order to match the volatility in the data, which further contributes to raise the computed cost (see appendix).

When hours are not held constant, their average value over the business cycle is slightly below steady state for RRACs ≥ 1, and marginally above for RRACs $\leq .5$. Average capital and consumption are both above their respective steady-state values, whether or not hours are constant, more so the greater the RRAC. Restricting hours to be constant makes the mean and standard deviation of

	α	β	δ	θ	ρ
Low	(.25)	(.985)	(.01)	(.5)	(.85)
	.0954	.0886	.2540	.1798	.1553
Baseline	(1/3)	(.99)	(.025)	(2)	(.9)
	.1185	.1185	.1185	.1185	.1185
High	(.45)	(.995)	(.04)	(3)	(.95)
	.1722	.1984	.0667	.1397	.1632

Table 3: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of steady-state consumption for different parameter values. MaCurdy preferences, RRAC = 2, $gp = 2\,000, gpz = 51$.

consumption increase for RRACs $\geq$ 2, while it makes both fall for RRACs $\leq$ 1. In particular, for a RRAC = 2, fixing hours at their steady-state level leads to a .27% increase in average hours, a .15% rise in average consumption, and a 15.7% increase in the (unfiltered) standard deviation of consumption. When the RRAC = 1, it leads to a .07% increase in average hours, a .16% decrease in average consumption, and a 19.3% drop in the standard deviation of consumption.[3]

The computed cost of keeping hours constant over the business cycle are not very sensitive to the parametrization of the model. Table 3 presents results varying the different parameter values, one by one, from the baseline, while keeping the RRAC equal to 2. For the values used, increasing the labor supply elasticity θ, decreasing capital depreciation δ, lowering persistence ρ, or increasing β or α, all raise the cost, though only marginally for given σ (largest is .16% for parameter values in the table). However, the described changes in α, β, δ and ρ all require raising the volatility σ of productivity to again make consumption and hours at least as volatile as in the data. This produces the somewhat higher costs reported in table 3. Lowering θ, increasing capital depreciation δ, raising persistence ρ, or decreasing β or α, all lower the cost of holding hours constant for unchanged σ, but most of these also require adjusting σ. Doing so makes the total effect from lowering θ or raising ρ be an increase in the computed cost for the suggested parameter values.

Reducing the labor supply elasticity θ decreases households' willingness to

[3]The reported statistics were calculated by running 101 million randomly generated productivity shocks z_t through the computed optimal decision rules and averaging over 1 000 repetitions. The simulations were all initiated at the non-stochastic steady state $z_0 = 0$ and $k_0 = k_{ss}$, but the first 1 million values were discarded prior to calculating statistics.

Pref	RRAC 0	.5	1	2	3	4	5	10	15	20
MaC	.0383	.0860	.0735	.1169	.2542	.5242	.9048	2.605	4.501	6.346
KPR	-	.0387	.0753	.0440	.0472	.0489	.0515	.0610	.0677	.0741
GHH	.0363	.0844	.0856	.0896	.0934	.0967	.0998	.1129	.1248	.1375

Table 4: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of steady-state consumption. MaCurdy, KPR and GHH preferences, $gp = 4\,000, gpz = 51$.

vary hours with productivity, thus lowering the welfare cost of forcing hours to be constant for any $\sigma > 0$. However, since the standard deviation of hours falls with θ, σ must be raised for it to again match that in the data. As a result, the overall effect is that the computed cost of constant hours increases when lowering the labor supply elasticity from 2 to .5, but it is still just .18%. For $\theta = 1$, the cost drops to .09%.

Lowering depreciation δ to .01 raises the cost of constant hours to .25% of consumption, the largest in table 3. However, it also reduces the ratio of the standard deviation of consumption to that of hours in the model, which is already lower than in the data for baseline parameter values, thus worsening the model fit in this respect (.66 in data, .46 in baseline model, .27 with $\delta = .01$, all HP-filtered). Adding on all the other parameter changes in table 3 that contribute to raising the cost of constant hours ($\alpha = .45$, $\beta = .995$, $\theta = 3$, $\rho = .85$), yields a cost of .47% of consumption, or $264 per capita per year in the U.S. in 2023, which is still relatively modest.

Table 4 shows the computed cost of maintaining hours constant over the business cycle for different utility functions and RRACs, assuming baseline parameter values.[4] For risk aversion of 2 and above, the cost is larger with MaCurdy preferences than with GHH or KPR utility. Part of the reason is that the σ required to match the volatility in the data is increasing in the RRAC for MaCurdy, but decreasing or constant in the RRAC for KPR and GHH (see appendix). For RRACs below 2, the three utility functions yield similar results.

There are multiple reasons why our computations may overestimate the welfare cost of eliminating cyclical fluctuations in hours. First and foremost, we assume all fluctuations are in terms of relatively innocuous intensive-margin

[4] With GHH preferences, the RRAC is endogenous, so only its steady-state value is exactly as reported in the table (see appendix for details).

adjustments, rather than costly layoffs and hiring. In fact, other than the impact on household utility from smoothing leisure over time, there is no direct benefit in our model from eliminating fluctuations in hours. Two, we use values for σ high enough for both consumption and hours to be at least as volatile as in the data, making the model economy more turbulent than reality overall, and the cost of constant hours is increasing in σ. Three, we assume that the only source of volatility is productivity, which raises the gains from varying hours, rather than shocks that would not directly impact labor productivity, such as those to government spending. Four, the optimal level of constant hours may not correspond to the steady-state value used in our computations. Finally, the computed cost is decreasing in the number of gridpoints for capital (see appendix), but we can only employ a finite number of these.

Everything else equal, allowing for a finer adjustment of hours would raise the computed gains from varying these with productivity, and thus the cost if they are instead held constant. Our grid for hours is linked to that for capital through the first-order condition (5), so the two factors of production have the same number of gridpoints. While this is a balanced approach, it nevertheless makes the cost of constant hours decrease with the number of gridpoints, though the impact is minor beyond 1 000 gridpoints (see appendix).

Ignoring hiring and firing costs, and other factors pushing employers to hoard labor over the business cycle, contribute to overestimate the cost of keeping aggregate hours constant in our computations, for any given σ. However, it can also affect the calibration of σ, especially if labor hoarding in practice also contributes to underreporting of fluctuations in hours. This, or any other modification to the model that would require increasing the standard deviation of productivity shocks σ to match the volatility in the data, could raise the computed cost of keeping hours constant, since it is increasing in σ. However, as the tables in the appendix show, the cost is not very sensitive to changes in σ for standard parameter values, including a RRAC between one and two. Going 50% above the currently calibrated values would still yield a computed cost of less than .36% of consumption across the three utility functions. Even doubling σ would at most yield a cost of .5%. Of course, there could also be omissions in the model that warrant using a lower σ, for example variable capital utilization. After all, the calibrated volatility of productivity shocks σ used is already unusually high.

4 Conclusions

Even ignoring costs associated with extensive-margin variations in aggregate hours, such as temporary unemployment, the loss of firm-specific capital, labor-market search and screening, the drop in welfare from keeping aggregate hours constant over the business cycle is at most the equivalent of .12% of consumption for standard parameter values. When pushing the calibration to the extreme, the impact remains below .5%. On the other hand, the constraint of constant hours is simply imposed in our model. In practice, the economy would have to be enticed to comply, which could introduce distortions and thus additional welfare costs. For example, some countries temporarily implemented wage subsidies during the COVID-19 crisis (see Eichhorst et al. (2022) for an overview). These encouraged businesses to keep their employees working full-time, rather than resorting to layoffs or furloughs, and facilitated this labor hoarding by providing liquidity. But such subsidies can also have detrimental effects by delaying structural change. Likewise, severance pay discourages temporary workforce reductions, but can stifle job creation.

5 References

Aiyagari, S. R. (1994), "On the Contribution of Technology Shocks to Business Cycles." *Federal Reserve Bank of Minneapolis Quarterly Review* 18 (3): 22-34.

Barro, R. J. (2009), "Rare Disasters, Asset Prices, and Welfare Costs." *American Economic Review* 99 (1): 243-64.

Becker, G. (1962), *Human Capital*, The University of Chicago Press.

Blatter, M., Muehlemann, S. and Schenker, S. (2012), "The costs of hiring skilled workers." *European Economic Review* 56: 20-35.

Boppart, T. and Krusell, P. (2020), "Labor Supply in the Past, Present, and Future: A Balanced-Growth Perspective." *Journal of Political Economy* 128 (1): 118-57.

Chetty, R. (2006), "A New Method of Estimating Risk Aversion." *American Economic Review* 96 (5): 1821-34.

Cho, J. and Cooley, T. F. (1994), "Employment and hours over the business cycle." *Journal of Economic Dynamics and Control*, 18 (2): 411-32.

Eichhorst, W., Marx, P., Rinne, U. and Brunner, J. (2022), "Job retention schemes during COVID-19: A review of policy responses." *IZA Policy Paper* 187.

Elminejad, A., Havranek, T. and Irsova, Z. (2022), "Relative Risk Aversion: A Meta-Analysis." *EconStor Preprints* 260586, ZBW - Leibniz Information Centre for Economics.

Fair, R. C. (1985), "Excess Labor and the Business Cycle." *American Economic Review* 75 (1): 239-45.

Fay, J. A. and Medoff, J. L. (1985), "Labor and Output Over the Business Cycle: Some Direct Evidence." *American Economic Review* 75 (4): 638-55.

Gomes, J., Greenwood, J. and Rebelo, S. (2001), "Equilibrium unemployment." *Journal of Monetary Economics* 48 (1): 109-52.

Greenwood, J., Hercowitz Z. and G. Huffman G. W. (1988), "Investment, Capacity Utilization, and the Real Business Cycle." *American Economic Review* 78 (3): 402-17.

Hodrick, R. and Prescott, E. C. (1997), "Postwar U.S. Business Cycles: An Empirical Investigation." *Journal of Money, Credit, and Banking* 29 (1): 1-16.

King, R. G., Plosser, C. I. and Rebelo, S. T. (1988), "Production, Growth and Business Cycles: I. The Basic Neoclassical Model." *Journal of Monetary Economics* 21 (2-3): 195-232.

Krueger, A. B. and Mueller, A. (2010), "Job search and unemployment insurance: New evidence from time use data." *Journal of Public Economics* 94 (3): 298-307.

Krusell, P., Mukoyama, T., Şahin, A. and Smith, A. A. (2009), "Revisiting the welfare effects of eliminating business cycles." *Review of Economic Dynamics* 12 (3): 393-404.

Krusell, P. and Smith, A. A. (1999), "On the Welfare Effects of Eliminating Business Cycles." *Review of Economic Dynamics* 2 (1): 245-72.

Kydland, F. E. and Prescott, E. C. (1982), "Time to Build and Aggregate Fluctuations." *Econometrica* 50 (6): 1345-70.

Lazear, E. P. (2009), "Firm-Specific Human Capital: A Skill-Weights Approach." *Journal of Political Economy* 117 (5): 914-40.

Ljungqvist, L. and Sargent, T. J. (2018), *Recursive Macroeconomic Theory*, MIT Press.

Lucas, R. E. (1987), *Models of Business Cycles*, Oxford: Basil Blackwell.

Lucas, R. E. (2003), "Macroeconomic Priorities." *American Economic Review* 93 (1): 1-14.

MaCurdy, T. E. (1981), “An Empirical Model of Labor Supply in a Life-Cycle Setting.” *Journal of Political Economy* 89 (6): 1059-85.

Miao, J. (2020), *Economic Dynamics in Discrete Time*, MIT Press.

Okun, A. (1963), “Potential GNP: Its Measurement and Significance.” *Cowles Foundation Paper* 190”.

Otrok, C. (2001), “On measuring the welfare cost of business cycles.” *Journal of Monetary Economics* 47 (1): 61-92.

Storesletten, K., Telmer, C. I. and Yaron, A (2001), “The Welfare Cost of Business Cycles Revisited: Finite Lives and Cyclical Variation in Idiosyncratic Risk.” *European Economic Review* 45 (7): 1311-39.

Tallarini, T. D. (2000), “Risk-Sensitive Real Business Cycles.” *Journal of Monetary Economics* 45 (3): 507-32.

Topel, R. (1990). “Specific capital and unemployment: Measuring the costs and consequences of job loss.” *Carnegie-Rochester Conference Series on Public Policy* 33: 181-214.

Trabandt, M. and Uhlig, H. (2011), “The Laffer Curve Revisited.” *Journal of Monetary Economics* 58: 305-27.

A Appendix (for online publication)

A.1 Gridpoints productivity shocks

The discretization of the stochastic shock z_t with the code *markovappr.m* makes its simulated persistence rise, and standard deviation fall, both marginally, as the number of gridpoints *gpz* is increased. At the same time, for any given width used for the discrete grid for z_t, raising the number of gridpoints eventually leads to bunching at the extremes. As a compromise between these two effects, and the computational resources required to solve the models, we use 51 gridpoints for z_t distributed symmetrically around its unconditional mean of zero up to four standard deviations in either direction. Figure A2 provides a graphical representation, showing the percentage distribution of 100 million random draws across the 51 gridpoints assuming $\sigma = 1$. Note that in our computations, the gridpoint locations vary with σ as determined by the algorithm *markovappr.m*. Table A1 shows how the welfare cost from eliminating the cyclical fluctuations in hours vary with the number of gridpoints *gpz* used in the computations for MaCurdy preferences. The numbers barely vary when *gpz* is greater than 31, but are slightly decreasing in *gpz*.

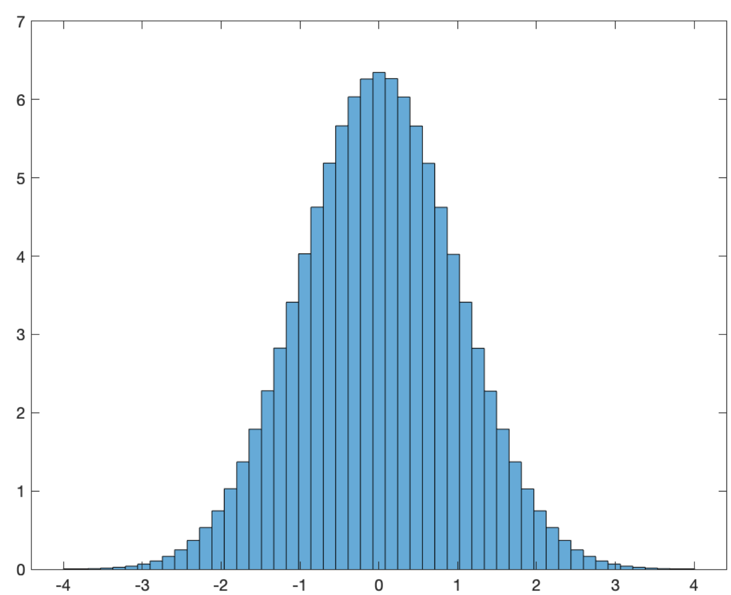

Figure A2: Sample distribution of z in simulations with 51 gridpoints, $\sigma = 1$.

	RRAC									
gpz	0	.5	1	2	3	4	5	10	15	20
101	.0380	.0855	.0735	.1175	.2548	.5239	.9027	2.596	4.484	6.323
91	.0380	.0855	.0735	.1176	.2550	.5242	.9033	2.598	4.487	6.327
81	.0381	.0856	.0736	.1177	.2552	.5247	.9041	2.600	4.492	6.335
71	.0381	.0857	.0737	.1179	.2556	.5254	.9054	2.604	4.499	6.344
61	.0382	.0859	.0738	.1181	.2561	.5265	.9074	2.610	4.508	6.358
51	.0383	.0862	.0741	.1185	.2569	.5285	.9103	2.618	4.523	6.378
41	.0386	.0867	.0745	.1192	.2585	.5315	.9159	2.635	4.553	6.420
31	.0391	.0879	.0755	.1208	.2619	.5385	.9282	2.671	4.616	6.509

Table A1: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with gpz gridpoints for z. MaCurdy preferences, $gp = 2\,000$.

A.2 Gridpoints capital

The grid for capital is constructed as follows for all our models. 32% of the gridpoints are equally spaced out (linearly) between zero and $k_{ss} = 100$. 64% are equally spaced out between k_{ss} and $3k_{ss}$. The remaining 4% of gridpoints are equally spaced out between $3k_{ss}$ and $13k_{ss}$. With 8 000 gridpoints, these are .039 units apart up to $3k_{ss}$, then 3.145 units apart between $3k_{ss}$ and $13k_{ss}$. With 4 000 gridpoints, these distances are twice as large, so .078 and 6.329, respectively. Our computational resources could at most handle 8 000 gridpoints with GHH utility, and 4 000 with KPR and MaCurdy preferences. But as the tables below show, the numbers barely change when going beyond 2 000 gridpoints.

While capital does not venture as far away from its steady-state value as our grid might suggest, it is important for the grid to be wide enough to include all values with a positive probability in the value-function iterations. At the same time, for a given number of gridpoints, a wider grid means there is more space between gridpoints, bringing us further away from the continuous capital choice in the original model. As a compromise, we increase the spacing between gridpoints for capital levels above $3k_{ss}$. We found results to be essentially the same with the finer grid stretching only up to $2k_{ss}$, and using a maximum value of $5k_{ss}$, once adjusting the total number of gridpoints so that the distance between these was unchanged.

A finer grid allows for a more precise adjustment of capital, and thus a higher maximum expected lifetime utility (1). The improvement turns out to be larger when hours worked are held constant, making the computed cost of constant hours fall with the number of gridpoints for capital, as is illustrated in table A2. This despite more gridpoints allowing for a finer adjustment of hours.

	RRAC									
gp	0	.5	1	2	3	4	5	10	15	20
4 000	.0383	.0860	.0735	.1169	.2542	.5242	.9048	2.605	4.501	6.346
2 000	.0383	.0862	.0741	.1185	.2569	.5285	.9103	2.618	4.523	6.378
1 000	.0383	.0870	.0765	.1249	.2680	.5441	.9319	2.669	4.609	6.503
500	.0383	.0903	.0864	.1507	.3121	.6081	1.018	2.880	4.952	6.986

Table A2: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with gp gridpoints for k. MaCurdy preferences, $gpz = 51$.

A.3 Calibrating volatility σ

The U.S. data used to calibrate volatility is real private consumption (PCECC96), hours worked for all workers in the nonfarm business sector (HOANBS), and real GDP (GDPC1), all downloaded from the FRED database on January 17th, 2025.[5] The three are seasonally adjusted quarterly series spanning 1947Q1 to 2023Q4. The HP-filtered standard deviation is 1.39 for consumption, 2.09 for hours and 1.64 for GDP, represented by dotted lines in the figures below, which include the equivalent values from the model economy plotted as functions of σ for different RRACs. As the figures illustrate, it is matching the standard deviation of consumption that requires the highest volatility σ for baseline parameter values when RRAC $\geq .5$ with MaCurdy preferences (hours for RRAC $= 0$). Note also that the required σ is increasing in the RRAC.

A.4 Cost for different levels of volatility σ

The computed cost of constant hours is increasing in the volatility σ of productivity shocks, as table A3 illustrates.

[5] https://fred.stlouisfed.org.

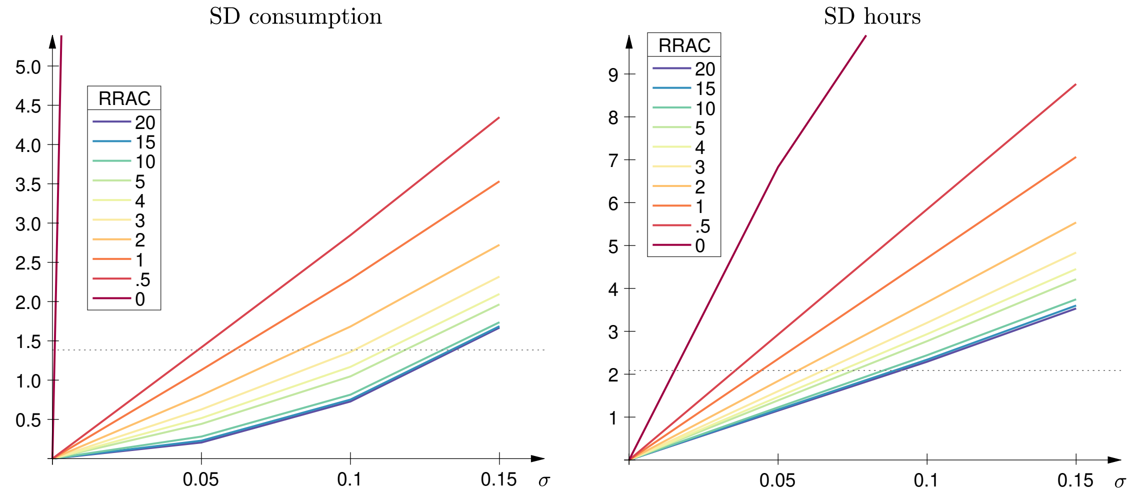

Figure A3: Simulated HP-filtered standard deviations for various levels of volatility σ. MaCurdy preferences.

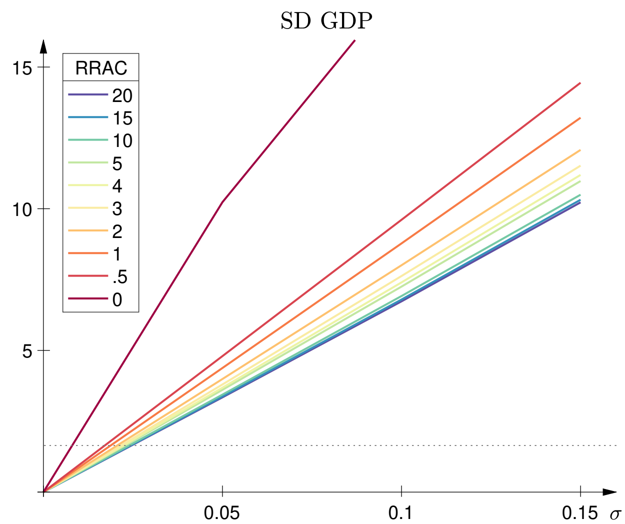

Figure A4: Simulated standard deviation for HP-filtered GDP for various levels of volatility σ. MaCurdy preferences.

	RRAC									
σ	0	.5	1	2	3	4	5	10	15	20
.05	.4001	.0910	.0494	.0405	.0627	.0956	.1331	.3368	.5430	.7474
.10	1.565	.3654	.1969	.1603	.2477	.3783	.5277	1.358	2.228	3.122
.15	3.556	.8282	.4416	.3592	.5546	.8479	1.186	3.117	5.203	7.333
.20	6.618	1.487	.7823	.6360	.9808	1.502	2.109	5.660	9.472	13.16

Table A3: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, for different levels of volatility σ. MaCurdy preferences, $gp = 4\,000, gpz = 51$.

A.5 Cost using $\bar{c}, \bar{k}$ instead of c_{ss}, k_{ss}

With uncertainty, the economy will deviate systematically from the non-stochastic steady state. In particular, average capital over the business will be higher than its non-stochastic steady-state value. One might therefore argue that the cost of keeping hours constant should be evaluated at the average capital stock pertaining when hours are free to vary, setting $k_0 = \bar{k}$ in $v(k_0, z_0)$, and as a fraction of the corresponding average consumption $\bar{c}$.[6] Table A4 does so. Comparing with table 4 shows that except for RRACs above 5 with MaCurdy preferences, the cost is virtually the same whether using average or steady-state values. For RRACs above 5 with MaCurdy preferences, the computed cost using averages is lower than with steady-state values.

	RRAC									
Pref	0	.5	1	2	3	4	5	10	15	20
MaC	.0382	.0858	.0735	.1179	.2570	.5279	.9028	2.458	4.068	5.626
KPR	-	.0387	.0753	.0441	.0474	.0491	.0518	.0618	.0695	.0781
GHH	.0363	.0842	.0855	.0896	.0935	.0969	.1000	.1128	.1244	.1377

Table A4: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of $\bar{c}$. MaCurdy, KPR and GHH preferences. $gp = 4\,000, gpz = 51$.

A.6 Model with KPR preferences

With King-Plosser-Rebelo (KPR henceforth, see King et al. (1988)) preferences,

$$u\left(c_t, n_t\right) = \frac{c_t^{1-\varepsilon}\left(1 - \frac{\psi(1-\varepsilon)}{1+1/\theta} n_t^{1+1/\theta}\right)^{\varepsilon} - 1}{1-\varepsilon} \tag{11}$$

the RRAC is ε, while the Frisch elasticity is θ (see Trabandt and Uhlig (2011)). The parameter $\psi > 0$. We use the budget constraint

$$c_t = (1 + z_t) A k_t^{\alpha} n_t^{1-\alpha} + (1-\delta) k_t - k_{t+1} \tag{12}$$

instead of the one above (3) in order to reuse computations from a different project, but this has a negligible impact of the results.[7] The rest of the model

[6] The distinction between average and steady state makes no difference for productivity, since $\bar{z} = z_{ss} = 0$.

[7] $E\left(\exp\left(z_t\right)\right) > \exp\left(E\left(z_t\right)\right)$ due to the convexity of the exponential function (see Jensen's inequality), making average productivity increase exogenously with volatility σ when using $\exp\left(z_t\right)$

remains unchanged.

The first-order condition for hours of work is

$$\varepsilon\psi c_t n_t^{1/\theta}\left(1-\frac{\psi\left(1-\varepsilon\right)}{1+1/\theta}n_t^{1+1/\theta}\right)^{-1}=\left(1-\alpha\right)\left(1+z_t\right)Ak_t^{\alpha}n_t^{-\alpha} \tag{13}$$

which we use to solve numerically for n_t as a function of z_t, k_t and k_{t+1}, and thus substitute it out of the grid search. As with MaCurdy preferences, we use 4 000 gridpoints for capital, and set the parameters A and ψ so that $k_{ss} = 100$ and $n_{ss} = 1/3$. All other free parameters remain unchanged, with $\alpha = 1/3, \beta = .99, \delta = .025, \rho = .9$ and $\theta = 2$. In this case, a change in expected lifetime utility Δv is equivalent to a change

$$x=\left[\left(1-\beta\right)\left(1-\varepsilon\right)\left(1-\frac{\psi\left(1-\varepsilon\right)}{1+1/\theta}n_{ss}^{1+1/\theta}\right)^{-\varepsilon}c_{ss}^{\varepsilon-1}\Delta v+1\right]^{\frac{1}{1-\varepsilon}}-1 \tag{14}$$

in steady-state consumption $c_{ss} = 8.03$.

As is shown in figure A5, for RRACs ≥ 1, the standard deviations of HP-filtered consumption and hours are increasing in the RRAC with KPR preferences. Hence, the larger the RRAC, the lower the σ required to match the volatility in the data, as is shown in table A5.

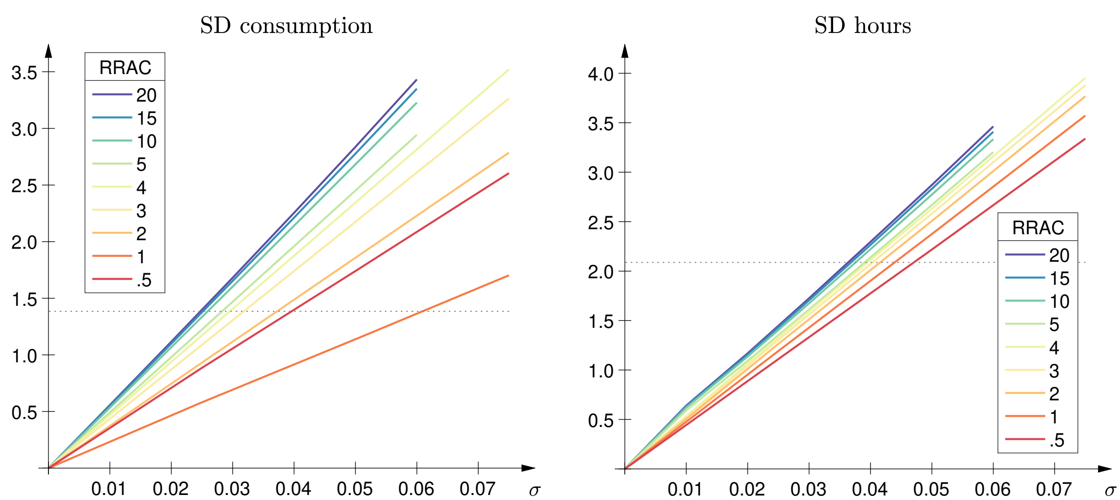

Figure A5: Simulated HP-filtered standard deviations for different levels of volatility σ. KPR preferences.

in production. With $1 + z_t$, increasing σ instead leads to a mean-preserving spread. The drawback is that there will be a non-zero probability of negative productivity $A(1 + z_t)$, but the values of σ used are low enough for this not to occur in the present study.

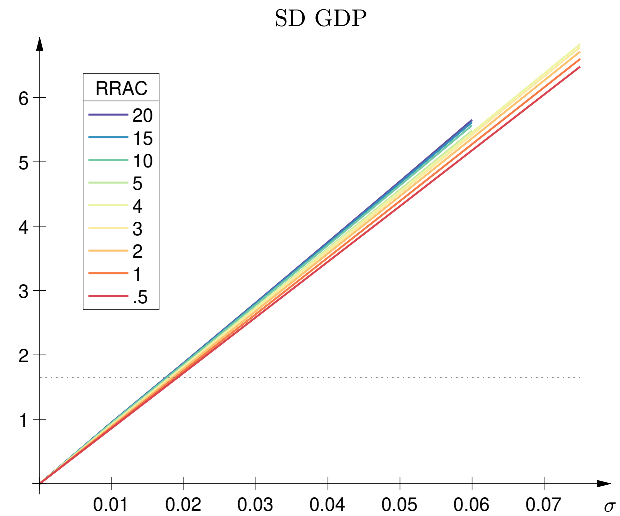

Figure A6: Simulated standard deviation of HP-filtered GDP for different levels of volatility σ. KPR preferences.

	RRAC								
	.5	1	2	3	4	5	10	15	20
σ	.0475	.06125	.0425	.04125	.0400	.0395	.0380	.0370	.0364

Table A5: Volatility σ necessary for simulated standard deviation of HP-filtered consumption and hours to be at least as large as in U.S. data. KPR preferences.

	RRAC								
gp	.5	1	2	3	4	5	10	15	20
4 000	.0387	.0753	.0440	.0472	.0489	.0515	.0610	.0677	.0741
2 000	.0391	.0759	.0448	.0481	.0499	.0525	.0623	.0691	.0754
1 000	.0407	.0783	.0481	.0518	.0538	.0566	.0668	.0742	.0819
500	.0473	.0881	.0610	.0662	.0692	.0727	.0857	.0964	.1094

Table A6: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with gp gridpoints for k. KPR preferences, $gpz = 51$.

	RRAC								
gpz	.5	1	2	3	4	5	10	15	20
101	.0388	.0753	.0445	.0478	.0495	.0521	.0617	.0684	.0749
91	.0388	.0753	.0445	.0478	.0495	.0521	.0617	.0684	.0750
81	.0389	.0754	.0445	.0478	.0496	.0522	.0618	.0685	.0751
71	.0389	.0755	.0446	.0479	.0496	.0522	.0618	.0686	.0751
61	.0390	.0756	.0447	.0480	.0497	.0523	.0619	.0687	.0754
51	.0391	.0759	.0448	.0481	.0499	.0525	.0623	.0691	.0754
41	.0394	.0764	.0452	.0485	.0502	.0528	.0625	.0694	.0761
31	.0399	.0775	.0457	.0491	.0509	.0535	.0634	.0704	.0770

Table A7: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with gpz gridpoints for z. KPR preferences, $gp = 2\,000$.

	RRAC								
σ	.5	1	2	3	4	5	10	15	20
.01	.0018	.0022	.0027	.0030	.0033	.0036	.0045	.0050	.0054
.02	.0070	.0082	.0099	.0113	.0124	.0133	.0168	.0192	.0211
.03	.0155	.0182	.0220	.0251	.0276	.0297	.0377	.0437	.0489
.04	.0275	.0321	.0390	.0444	.0489	.0528	.0678	.0798	.0913
.05	.0429	.0502	.0609	.0695	.0766	.0828	.1075	.1299	.1539
.06	.0618	.0722	.0878	.1003	.1108	.1201	.1590	.1964	.2403
.10	.1722	.2018	.2469	.2841	.3174	.3486	.5109	.7441	1.141
.15	.3914	.4610	.5706	.6683	.7632	.8619	1.581	3.288	6.944
.20	.7067	.8382	1.057	1.271	1.503	1.776	4.860	13.91	24.60

Table A8: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, for different levels of volatility σ. KPR preferences, $gp = 4\,000, gpz = 51$.

A.7 Model with GHH preferences

With Greenwood-Hercowitz-Huffman (GHH henceforth, see Greenwood et al. (1988)) utility,

$$u(c_t, n_t) = \frac{\left(c_t - \frac{\psi}{\omega} n_t^{\omega}\right)^{1-\varepsilon} - 1}{1-\varepsilon} \tag{15}$$

the Frisch labor supply elasticity is $1/(\omega - 1)$, where $\omega > 0$. The RRAC for consumption is endogenous and equal to $\varepsilon \times c_t/(c_t - \psi n_t^{\omega}/\omega)$, but we show below that its average value over the business cycle does not vary much in our simulations. We refer to its steady-state value $\varepsilon \times c_{ss}/(c_{ss} - \psi n_{ss}^{\omega}/\omega)$ as SSRRAC. Otherwise, the model remains as with KPR preferences.

Using the first-order condition for hours of work, we can solve for the optimal level as a function of capital k_t and productivity z_t,

$$n_t = \left(\psi^{-1}(1-\alpha)(1+z_t) A k_t^{\alpha}\right)^{\frac{1}{\omega+\alpha-1}} \tag{16}$$

and substitute out for n_t. Having an explicit solution saves on computational resources, allowing us to use up to 8 000 gridpoints for k. As above, A and ψ are set so that $k_{ss} = 100$ and $n_{ss} = 1/3$. All other free parameters have the same baseline values as above, that is, $\alpha = 1/3, \beta = .99, \delta = .025, \rho = .9$, while $\omega = 1.5$ so that the labor supply elasticity is again 2. A change in expected lifetime utility Δv is now equivalent to a change

$$x = c_{ss}^{-1}\left[\left((1-\beta)(1-\varepsilon)\Delta v + \left(c_{ss} - \psi\omega^{-1} n_{ss}^{\omega}\right)^{1-\varepsilon}\right)^{\frac{1}{1-\varepsilon}} + \psi\omega^{-1} n_{ss}^{\omega}\right] - 1 \tag{17}$$

in steady-state consumption $c_{ss} = 8.03$.

As figure A7 illustrates, matching the standard deviation in HP-filtered hours requires a higher σ than matching that of consumption. The standard deviation of hours barely varies with the SSRRAC, so the same value of $\sigma = .0325$ is used for all SSRRAC $\geq .5$, as is reflected in table A9. Values of σ greater than .05 are incompatible with keeping hours constant at their steady-state value since there would then be a non-zero probability of $c_t - \psi n_{ss}^{\omega}/\omega$ being negative, yielding an imaginary value for utility when $\varepsilon > 0$.

For our baseline parameter values, the steady-state value of $c_t/(c_t - \psi n_t^{\omega}/\omega)$ equals 2.397, which we use to adjust ε to yield the desired SSRRAC. With variable work effort, $\bar{c}/(\bar{c} - \psi\bar{n}^{\theta}/\theta)$ is decreasing in σ for SSRRAC $\in [.5, 5]$, and increasing for SSRRAC $\in [10, 20]$, but remains between 2.393 and 2.419

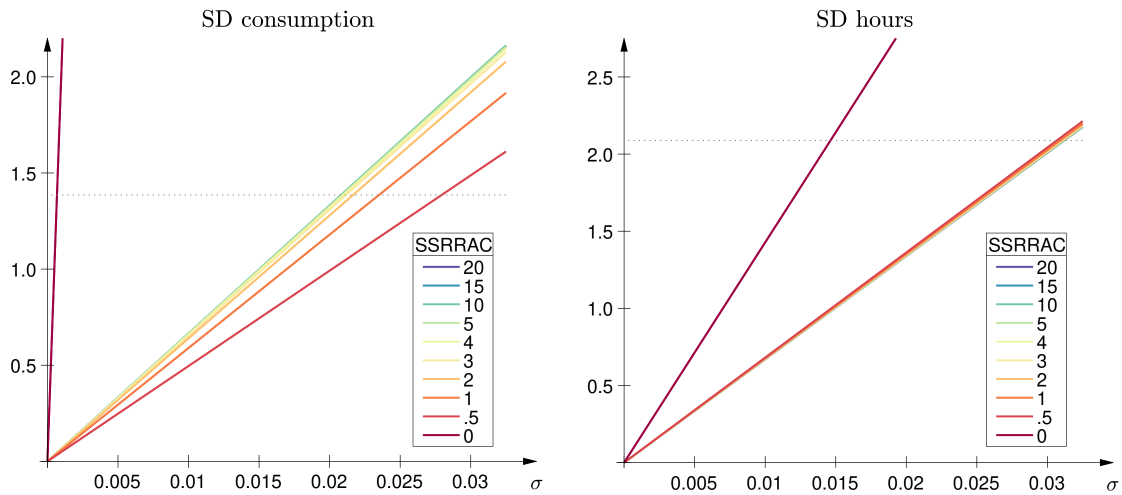

Figure A7: Simulated HP-filtered standard deviations for different levels of volatility σ. GHH preferences.

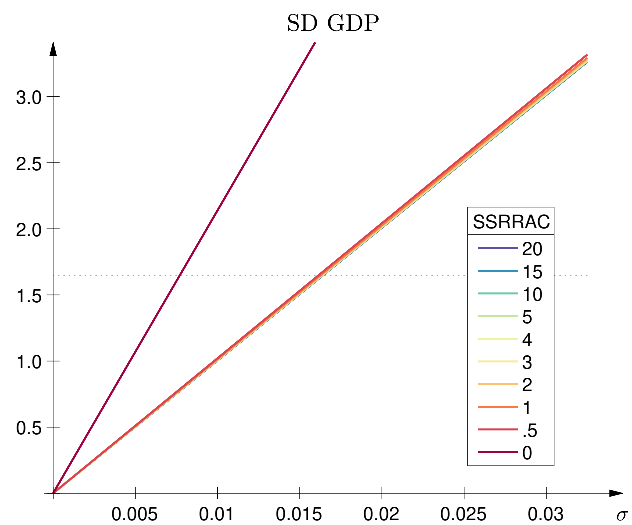

Figure A8: Simulated standard deviation of HP-filtered GDP for different levels of volatility σ. GHH preferences.

	SSRRAC									
	0	.5	1	2	3	4	5	10	15	20
σ	.0150	.0325	.0325	.0325	.0325	.0325	.0325	.0325	.0325	.0325

Table A9: Volatility σ necessary for simulated standard deviation of HP-filtered consumption and work effort to be at least as large as in U.S. data. GHH preferences.

for $\sigma \leq .0325$. With hours constant at n_{ss}, $\bar{c}/(\bar{c} - \psi n_{ss}^{\theta}/\theta)$ is decreasing in σ for SSRRAC $\in [.5, 20]$, but at most falls to 2.379 for $\sigma \leq .0325$. For SSRRAC $\in [.5, 4]$, the average RRAC is at least as high with constant hours n_{ss} as with variable hours, though at most .15% higher for $\sigma \leq .0325$. For SSRRAC $\in [5, 20]$, the average RRAC is lower with constant hours, but for $\sigma \leq .0325$ by at most .01% for SSRRAC = 5, .4% for SSRRAC = 10, 1% for SSRRAC = 15, and 1.6% for SSRRAC = 20. Hence, the average RRAC over the business cycle is almost indistinguishable between the different cases for SSRRAC ≤ 10, but slightly higher with constant hours for SSRRAC $\in [.5, 4]$.

	SSRRAC									
gp	0	.5	1	2	3	4	5	10	15	20
8 000	.0363	.0844	.0856	.0896	.0933	.0967	.0998	.1128	.1246	.1370
4 000	.0363	.0844	.0856	.0896	.0934	.0967	.0998	.1129	.1248	.1375
2 000	.0363	.0844	.0856	.0897	.0934	.0968	.0999	.1131	.1254	.1385
1 000	.0363	.0845	.0857	.0897	.0936	.0970	.1002	.1140	.1272	.1416
500	.0363	.0842	.0851	.0889	.0928	.0965	.0999	.1161	.1336	.1549

Table A10: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with *gp* gridpoints for k. GHH preferences, $gpz = 51$.

	SSRRAC									
gpz	0	.5	1	2	3	4	5	10	15	20
101	-	.0837	.0849	.0889	.0926	.0960	.0990	.1121	.1242	.1370
91	-	.0838	.0851	.0891	.0929	.0964	.0996	.1135	.1267	.1409
81	-	.0839	.0851	.0892	.0930	.0965	.0997	.1136	.1267	.1411
71	-	.0840	.0852	.0893	.0932	.0966	.0998	.1137	.1269	.1414
61	-	.0842	.0854	.0895	.0933	.0968	.1000	.1139	.1273	.1421
51	.0363	.0844	.0856	.0897	.0934	.0968	.0999	.1131	.1254	.1385
41	-	.0849	.0862	.0902	.0940	.0974	.1005	.1138	.1261	.1392
31	-	.0861	.0874	.0914	.0952	.0986	.1018	.1151	.1276	.1414

Table A11: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, computed with *gpz* gridpoints for z. GHH preferences, $gpz = 2\,000$.

A.8 Code

The MATLAB files used to compute all results in the present paper are available at https://github.com/cjensenc/rbcn. The files exploit parallel computing,

σ	SSRRAC 0	.5	1	2	3	4	5	10	15	20
.01	.0161	.0080	.0081	.0085	.0088	.0091	.0094	.0104	.0111	.0117
.02	.0645	.0320	.0324	.0339	.0353	.0365	.0376	.0419	.0453	.0483
.03	.1447	.0719	.0730	.0763	.0795	.0823	.0849	.0957	.1051	.1147
.04	.2552	.1279	.1297	.1358	.1416	.1469	.1518	.1736	.1956	.2207
.05	.3947	.1999	.2028	.2125	.2219	.2306	.2389	.2789	.3244	.3824

Table A12: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of c_{ss}, for different levels of volatility σ. GHH preferences, $gp = 8\,000, gpz = 51$.

with *cpus* denoting the number of CPUs available for the code to run on. Ideally, this should be equal to the product of the size of the vector containing the ε-values and *sdzsteps*+1. Note that parallel computing is RAM intensive. Runs with $8\,000 \times 51$ gridpoints were completed on a system with 2 015 GB RAM over 2-6 days. However, the results in the paper can be replicated on most personal computers with the following adjustments to speed up computations or lower RAM requirements:

- reduce the number of gridpoints for k, denoted *gp* in code. Results for 4 000, 2 000, 1 000 and 500 are reported in this appendix.
- reduce the number of different values for volatility σ computed in each run, denoted *sdzsteps* in code. The minimum is one.
- reduce the number of different values for ε, denoted *epsilonvect* in the code. The minimum is one value.
- lower the length of each simulation and the number of simulations, T and *Reps*, respectively, in the code.
- comment out HP-filtering in simulations, especially if T or *Reps* large.

Parallel computing can be disabled to save on RAM by replacing *parfor* in the code with *for*, and commenting out the *parpool* command.